The above data refer to which one of the 'Population by Religion' Groups?

(a) Sikhs (b) Jains
(c) Christians (d) Buddhists

Ans. (c) : The given data refers to the decadal population of Christians in India. In 2001 it was 24.08 million and in census 2011 it was 2.78 crores.

145. A worker reaches his factory 3 minutes late if his speed from his house to the factory is 5km/hr. If he walks at a speed of 6 km/hr, then he reaches the factory 7 minutes early. The distance of the factory from his house is:

(a) 4 km (b) 5 km
(c) 6 km (d) 7 km

Ans. (b) : Let the distance be x km then

$$\frac{x}{5}-\frac{3}{60}=\frac{x}{6}+\frac{7}{60}\Rightarrow\frac{x}{5}-\frac{x}{6}=\frac{10}{60}=\frac{1}{6}$$

$$\frac{x}{30}=\frac{1}{6}\Rightarrow x=\frac{30}{6}=5\text{km.}$$

146. Two wires have their lengths, diameters and resistivities, all in the ratio of 1 : 2. If the resistance of the thinner wire is 10 ohms, the resistance of the thicker wire is:

(a) 5 ohms (b) 10 ohms
(c) 20 ohms (d) 40 ohms

Ans. (b) : Resistance of the wire $= R = \rho\frac{l}{A}$

where $A = \frac{rd^2}{4}$

$$\frac{R_2}{R_1}=\frac{\rho_2 l_2 d_1^2}{\rho_1 l_1 d_2^2}$$

$$\frac{R_2}{10}=\frac{1}{2}\times\frac{1}{2}\times\frac{2^2}{1^2}=1$$

$$R_2 = 10\Omega$$

147. Consider the following organizations

1. Atomic Minerals Directorate for Research & Exploration
2. Heavy Water Board
3. Indian Rare Earths Limited
4. Uranium Corporation of India

Which of these/are under the Department of Atomic Energy?

(a) 1 only (b) 1 and 4
(c) 2, 3 and 4 (d) 1, 2, 3 and 4

Ans. (d) : The Department of Atomic Energy constituted on 3 August 1954 is involved in the development of nuclear power technology, radiation, use of technology in agriculture, medicine and industry, and basic research. All four of the above options are related to the Department of Atomic Energy.

148. Consider the following statements about the minorities in India:

1. The Government of India has notified five communities, namely, Muslims, Sikhs, Christians, Buddists and Zoroastrians as minorities.
2. The National Commission for Minorities was given statutory status in 1993.
3. The smallest religious minority in India are the Zoroastrians.
4. The Constitution of India recognizes and protects religious and linguistic minorities.

Which of these statements are correct?

(a) 2 and 3 (b) 1 and 4
(c) 2, 3 and 4 (d) 1, 2, 3 and 4

Ans. (d) : The Indian Constitution recognizes and protects minorities on religious and linguistic grounds.

In 1993, the National Minorities Commission was given constitutional status. The Chairman and members of the Minorities Commission are appointed by the President. Parsi community is the most talented but least in population in India. When the question was asked there are 6 minority communities, namely Muslims (14.2%), Christians (2.3%), Sikhs (1.7%), Buddists (0.7%), Jains (0.4%) and Paris (0.006%).

149. A conveyer belt delivers baggage at the rate of 3 tons in 5 minutes, and a second conveyer belt delivers baggage at the rate of 1 ton in 2 minutes. How much time will it take to get 33 tons of baggage delivered, using both the conveyer belts?

(a) 25 minutes and 30 seconds
(b) 30 minutes
(c) 35 minutes
(d) 40 minutes and 45 seconds

Ans. (b) : Baggage delivered by first belt in 1 minute

$$=\left(\frac{3}{5}\right)\text{tons}$$

Baggage delivered by second belt in 1 minute $=\frac{1}{2}$ tons

Both deliver baggage in 1 minute

$$=\left(\frac{3}{5}+\frac{1}{2}\right)\text{tons}=\frac{11}{10}\text{tons}$$

$\therefore$ Required time

$$\frac{33}{\frac{11}{10}}\Rightarrow 33\times\frac{10}{11}=30\text{ minutes.}$$

150. Water is filled in a container in such a manner that its volume doubles after every five minutes. If it takes 30 minutes for the container to be full, in how much time will it be one-fourth full?

(a) 7 minutes (b) 10 minutes
(c) 20 minutes (d) 25 minutes

Ans. (c) : Take 1 litre for first 5 minutes. Then it becomes double afterwards every five minutes. We now get, 1, 2, 4, 8, 16, 32 for 5 mins, 10 mins, 15 mins, 20 mins, 25 mins, 30 mins. Total litres we get is "32" and 1/4th of this will be "8" which occurs at 20 mins.

UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2000

GENERAL STUDIES : PAPER-I

Time: 2 hours **Maximum Number: 200**

ANCIENT HISTORY

1. Assertion A : The origin of Feudal system in ancient India can be traced to military campaigns.
Reason R : There was considerable expansion of the Feudal system during the Gupta period.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (b) : The origin of feudal system in ancient India can be traced to military campaigns because the feudal lords provided military help to the king during the war. There was considerable expansion of feudal system during Gupta period.

2. Assertion A : Ashoka annexed Kalinga to the Mauryan Empire.
Reason R : Kalinga controlled land and sea routes to South India.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (a) : Ashoka annexed kalinga to the Mauryan empire in 261 BC. Though Ashoka abandoned the policy of Physical occupation in favour of Policy of Cultural conquest after the Kalinga War, yet he retained Kalinga after its conquest and incorporated it into his empire as Kalinga controlled land and sea routes to South India. Thus R explains A.

3. Assertion A : The emphasis of Jainism on non-'violence (ahimsa) prevented agriculturalists from embracing Jainism.
Reason R : Cultivation involved killing of insects and, pests.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (a) : Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.

4. Assertion A : The Aham and Puram Poems of the Padinen Kilukanakku group formed a continuation of the Sangam composition.
Reason R : They were included under the Post-Sangam works as against the Sangam works proper.

(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (a) : On the basis of interpretation and context, the Sangam literature can be described into two types viz. Aham (inner) and Puram (outer). The topics of Aham are related to personal and human aspects such as love and sexual things. The topics of Puram are related to human experiences and emotions such as Heroism, Valor, Ethics and Philanthropy. The poems have also been classified on nature themes which are known as ThinatPatinen. kilkanakku is the post Sangam work that is of Aham as well as Puram context.

5. The practice of military governorship was first introduced in India by the:

(a) Greeks (b) Shakas
(c) Parthians (d) Mughals

Ans. (a) : Greeks first introduced military governorship in India. They appointed their Governors called strategos.

6. Which one of the followings dynasties was ruling over North India at the time of Alexander's invasion?

(a) Nanda (b) Maurya
(c) Sunga (d) Kanva

Ans. (a) : Alexander invaded India in 326 BC. Nandas were ruling over India at that time. The chronological order of these 4 dynasties are: Nanda, Maurya, Sunga and Kanvas. The Nanda dynasty ruled in northern part of the Indian subcontinent during the 4th century BCE, and possibly during the 5th century BCE. The Nandas overthrew the Shishunaga dynasty in the Magadha region of eastern India, and expanded their empire to include a larger part of northern India.